

CAPM

Capital Asset Pricing Model

Assumptions of Market Efficiency

- Investors employ Markowitz Portfolio Theory to determine the efficient portfolio.
- Market Portfolio is the aggregate of all portfolios. If investors combine two or more portfolios on the minimum variance set, they get another portfolio on the minimum variance set.
- Based on individual risk aversion, each of them invests in one of the portfolios in the efficient frontier.

Assumption of Risk Perception

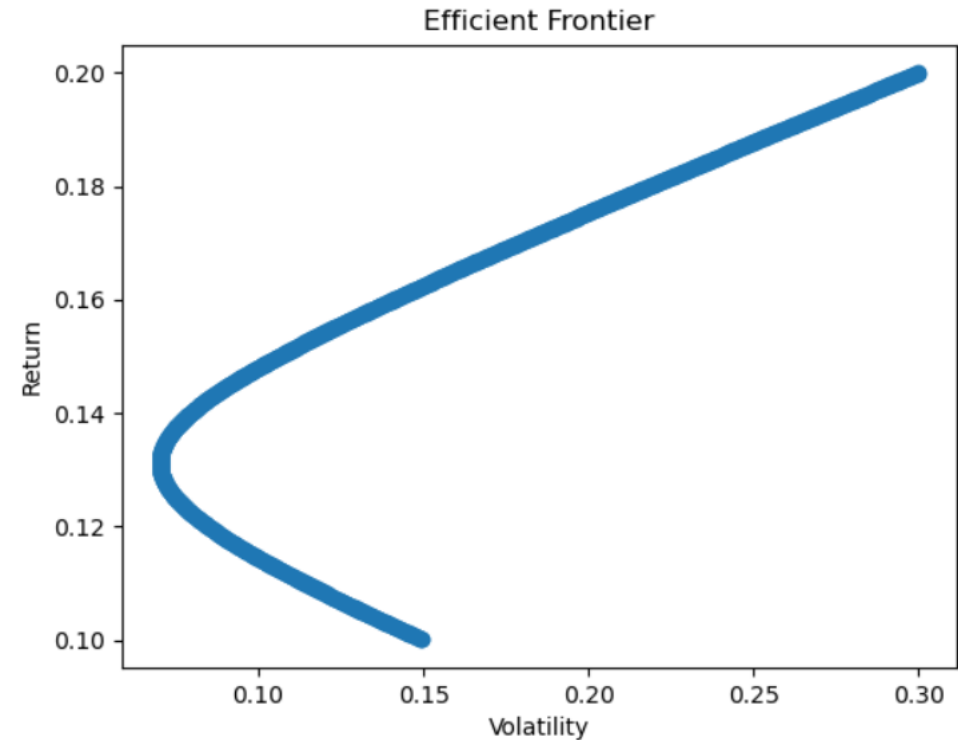
- Investors can choose between portfolios on the basis of expected return and variance.
 - If probability distributions for portfolio returns are all normally distributed, or
 - If investors' utility functions are all in quadratic form.
- If data is normally distributed, only two parameters are relevant: expected return and variance.
- If utility functions are quadratic, then relevant investment decision parameters are expected return and variance, even if other parameters exist.

Model Assumptions

- Returns on individual shares may be normally distributed (monthly) and positively skewed with limited loss and theoretically unlimited upside gains (yearly).
- Returns on portfolios may be normally distributed even if returns on individual shares are skewed.
- Investors have same planning and distributions of security returns.
- Market has no taxes, no transaction costs, no restrictions on short-selling efficient market.
- Assumptions are made in order to generate a model that examines the relationship between risk and expected return holding while keeping other factors constant.

Unlimited Borrowing and Lending

- Markowitz provides the covariance model to generate an efficient frontier based on all risky assets.
- Market provides unlimited borrowing and lending at risk free rate of return.



Capital Market Line

- Market line that goes through the risk-free rate of return and is tangent to the Markowitz efficient frontier.
- Point of tangency identifies the Market Portfolio (M).

- $$E(r_P) = r_F + \left[\frac{E(r_m) - r_F}{\sigma(r_m)} \right] \sigma(r_P)$$

Characteristic Line

- Systematic risk is measured statistically (OLS).
- Linear regression model measures both systematic risk and unsystematic risk.
- $r_i = \alpha_i + \beta_i r_m + \varepsilon_i$
 - α_i = Intercept of the i^{th} asset; β_i = slope (measure of systematic risk)
 - ε_i = Random error around the regression line for asset i .
- Regression relationship of the security with the market (Market Model)
- Regression model is formulated such that the error term averages out to zero.

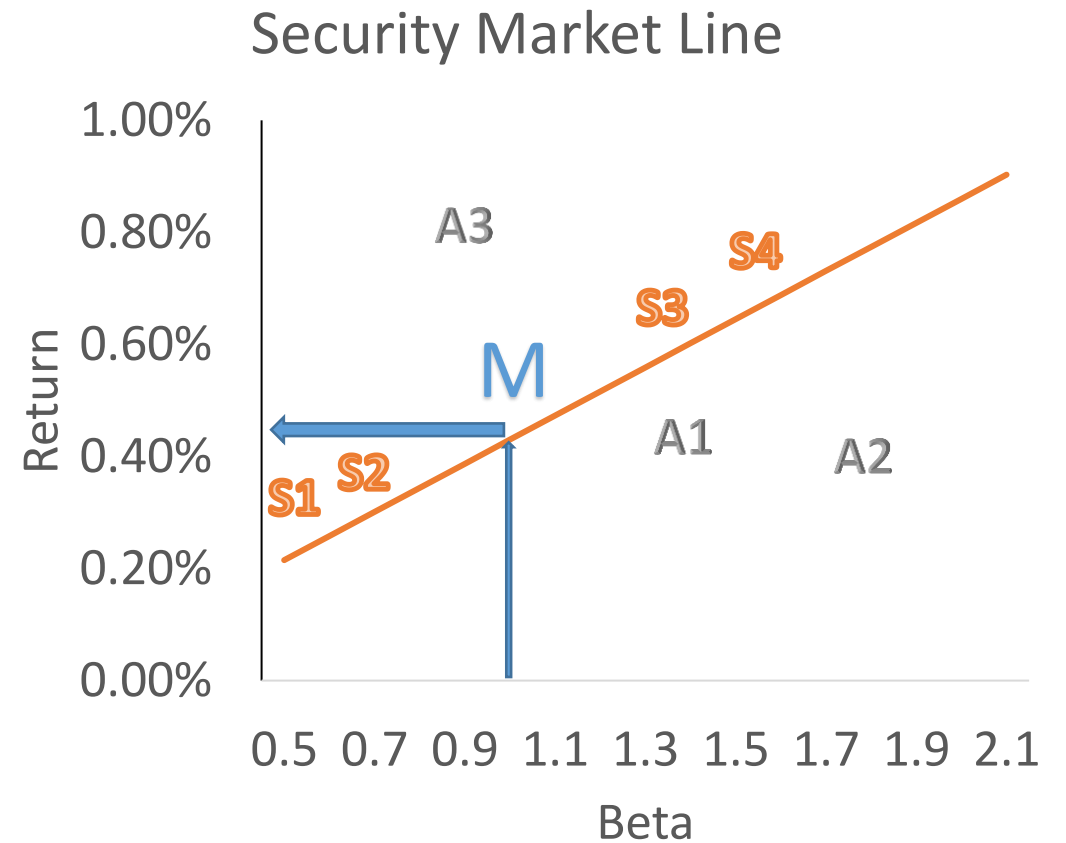
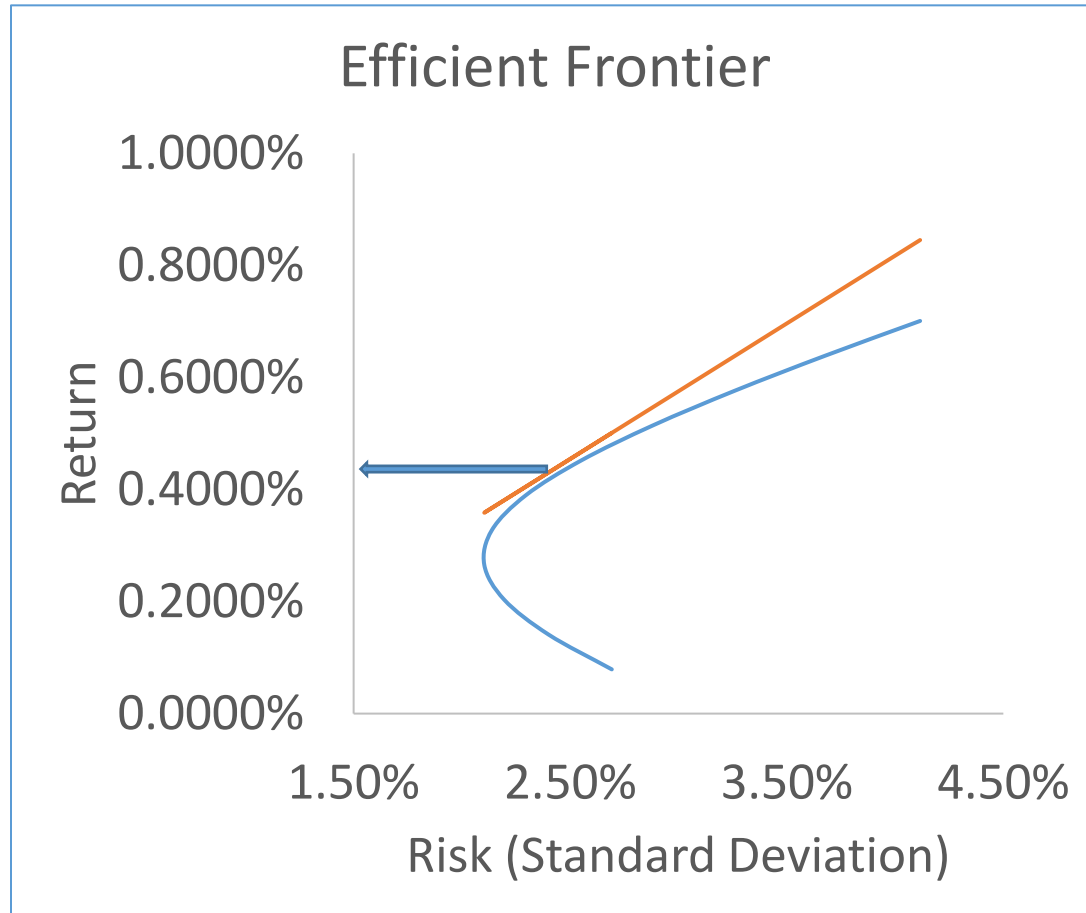
Security Market Line

- Given a population of securities, there will be a simple linear relationship between the beta factors of different securities and their expected (or average) returns if and only if the betas are computed using a minimum variance market index portfolio.
- Given the CML, SML can be determined (relationship between beta and expected return).
- $E(r_P) = r_F + [E(r_M) - r_F] \frac{\sigma(r_P)}{\sigma(r_M)}$
- $E(r_P) = r_F + [E(r_M) - r_F] \beta_P$ (Security Market Line)

CML Vs. SML

- CML
$$E(r_P) = r_F + \left[\frac{E(r_m) - r_F}{\sigma(r_m)} \right] \sigma(r_P)$$
- Restated
$$E(r_P) = r_F + [E(r_m) - r_F] \frac{\sigma(r_P)}{\sigma(r_m)}$$
- Portfolios on CML
$$\sigma(r_P) = \beta_P \sigma(r_M) \quad \beta_P = \frac{\sigma(r_P)}{\sigma(r_M)}$$

CML Vs. SML



Securities on CML

- Total risk $\sigma^2(r_P) = \beta_P^2 \sigma^2(r_M) + \sigma^2(\varepsilon_P)$

- Assumption $\sigma^2(\varepsilon_P) = 0$ for Diversified Portfolios

- Risk $\sigma^2(r_P) = \beta_P^2 \sigma^2(r_M)$ since $\beta_P = \sum_j^m x_j \beta_j$

- Security's contribution to the risk of a portfolio is measured by its beta. Since an individual security's residual variance can be diversified away in a portfolio, the market will not reward this "unnecessary" risk. Since only beta is relevant, individual securities will be priced to lie on the SML.

$$\sigma^2(r_P) = \left[\sum_j^m x_j \beta_j \right]^2 \sigma^2(r_M)$$

Application of CAPM

- Beta measures the responsiveness of a security to movements in the market portfolio.

$$\beta_i = \frac{Cov(R_i, R_M)}{\sigma^2(R_M)}$$

- Slope of the regression line is $\frac{\text{units of rise}}{\text{units of run}}$
- Beta will depend upon the choice of a proxy for the market portfolio.

CAPM

- Market Expected return $\bar{R}_M = R_F + \text{Market Risk Premium}$

- Security Expected return $\bar{R}_i = R_F + [\beta_i \times (\bar{R}_M - R_F)]$

$$\begin{array}{ccccccc} \text{Expected} & & & & & & \\ \text{return on a} & & & & & & \\ \text{security} & = & \text{Risk-free} & + & \text{Beta of the} & \times & \text{Market risk} \\ & & \text{rate} & & \text{security} & & \text{premium} \end{array}$$

- Assume $\beta_i = 0$, then the expected return is R_F .
- Assume $\beta_i = 1$, then $\bar{R}_i = \bar{R}_M$

Asset Beta

- Equity (Levered) beta = Asset (unlevered) beta x adjustment for debt holding in the firm

$$\beta_{\text{equity}} = \beta_{\text{asset}} \left[1 + (1 - \text{Tax rate}) \frac{\text{Debt}}{\text{Equity}} \right]$$

- Asset beta = Levered Equity beta
- High financial leverage increases equity beta

Levered & Unlevered Beta (US)

Industry Name	Number of firms	Beta	D/E Ratio	Effective Tax rate	Unlevered beta	Cash/Firm value	Unlevered beta corrected for cash	HiLo Risk	Standard deviation of equity	Standard deviation in operating income (last 10 years)
Advertising	54	1.34	26.20%	7.67%	1.12	6.56%	1.2	0.6098	67.19%	12.10%
Aerospace/Defense	67	0.9	22.79%	11.02%	0.77	3.87%	0.8	0.4341	42.74%	23.92%
Air Transport	24	1.24	106.83%	10.15%	0.69	10.28%	0.76	0.5255	65.27%	204.02%
Apparel	37	0.99	45.89%	8.08%	0.74	10.86%	0.83	0.4841	50.93%	24.49%
Auto & Truck	34	1.62	22.40%	2.11%	1.39	3.51%	1.43	0.6646	77.59%	46.93%
Auto Parts	33	1.23	47.83%	12.77%	0.91	8.61%	0.99	0.3843	52.62%	20.56%
Bank (Money Center)	15	0.88	183.19%	18.10%	0.37	29.77%	0.53	0.2337	30.47%	NA
Banks (Regional)	591	0.52	60.32%	16.84%	0.36	24.75%	0.48	0.2078	29.96%	265.19%
Beverage (Alcoholic)	18	0.61	30.47%	8.28%	0.5	1.86%	0.51	0.5509	63.08%	21.67%
Beverage (Soft)	29	0.57	19.73%	6.94%	0.5	3.57%	0.51	0.6427	47.67%	15.38%

Source: https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/Betas.html

Tax Adjusted CAPM

- Expected return

$$R_i = R_f(1 - T) + \beta_i[R_m - R_f - T(D_m - R_f)] + TD_i$$

where

T is the net adjusted tax rate and is defined as $(T_d - T_c)/(1 - T_c)$

T_d = tax rate on dividend

T_c = tax rate on capital gains

R_i = the expected return on a security i

R_f = risk free rate

β_i = beta of the security i

R_m = market return

D_m = dividend yield on market portfolio (index)

D_i = dividend yield on security i

Brennan-Lally CAPM

- Tax-adjusted CAPM was provided by Brennan (1970), whilst Lally (1992), amongst others, adjusted it for dividend imputation.
- Impetus for adoption of the tax-adjusted CAPM justifies the introduction of dividend imputation on firms, along with other aspects of the tax system (capital gains tax).
- Both the standard CAPM and the tax-adjusted-CAPM assume internationally segregated capital markets.

Comparison of CAPM and Tax Adjusted CAPM

- Brennan-Lally CAPM is an after-investor-tax model.
- Theoretical and application issues related to the model:
 - Average personal tax rate of investors must be estimated.
 - Model assumes that the economy is closed to international investors.

Comparison Parameter	CAPM	Tax Adjusted CAPM
Risk free Rate	Yield on government bonds	Yield on government bonds, after tax & risk-free rate (investor tax rate)
Beta	Equity beta	Equity beta
Risk Premium	Equity Market Return minus the risk-free rate	Tax adjusted EMRP (Equity Market Return minus the tax adjusted risk-free rate)

Firm's Market Valuation / Cost of Capital

- Theoretical Market valuation
 - If traded market price is higher than theoretical expectation firm's traded price is overvalued.
 - If traded market price is lower than theoretical expectation firm's market trade is undervalued.
- Arbitrage Opportunities
 - Overvalued securities create selling pressure in the market
 - Undervalued securities result in buying pressure in the market
- Expected return implies cost of owner's funds in the market.